

```
#include <Servo.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>


// Define objects

Servo gateServo;

LiquidCrystal_I2C lcd(0x27, 16, 2); // LCD I2C address (0x27 or 0x3F)


// Ultrasonic Sensor

const int trigPin = 8;

const int echoPin = 9;


// Variables

long duration;

int distance;


void setup() {

    // Servo setup

    gateServo.attach(6); // Servo motor connected to pin 6

    gateServo.write(0); // Gate initially closed


    // LCD setup

    lcd.init();

    lcd.backlight();

    lcd.setCursor(0, 0);

    lcd.print("Smart Gate Sys");

    delay(2000);

    lcd.clear();
```

```
// Ultrasonic setup
pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);
}

void loop() {
    // Get distance from Ultrasonic sensor
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);

    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    duration = pulseIn(echoPin, HIGH);
    distance = duration * 0.034 / 2; // Convert to cm

    lcd.setCursor(0, 0);
    lcd.print("Dist: ");
    lcd.print(distance);
    lcd.print(" cm ");

    // Gate Control Logic
    if (distance <= 25) { // If object detected within 25cm
        gateServo.write(90); // Open gate
        lcd.setCursor(0, 1);
        lcd.print("Gate: OPEN ");
    }
}
```

```
}  
else {  
    gateServo.write(0); // Close  
    lcd.setCursor(0, 1);  
    lcd.print("Gate: CLOSED");  
}  
  
delay(1000); // Wait 1 second before next measurement  
}
```